Quality Management

Chapter 3: The New Seven QC Tools
(7 Management & Planning Tools)

A Comprehensive Study Guide

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Prepared for exam revision & deep understanding

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1. Introduction — Why New Seven QC Tools?

1.1 The Need

The original 7 QC Tools (Histograms, Run Charts, Pareto Charts, etc.) are excellent for analyzing **numerical/quantitative data**. But in real-world management, many problems involve **verbal/qualitative data** — ideas, opinions, issues, strategies — that cannot be easily reduced to numbers.

The Japanese felt that managers should not only know the elementary statistical system but also **sophisticated techniques** for handling verbal and non-numerical information. The practice of TQC (Total Quality Control) requires not only numerical data but also **verbal data or verbal communication**, which may not be directly computable.

There needed to be suitable methods to process this verbal information with a view to arriving at a **complete plan**.

1.2 History

Year	Event
1972	A committee of **J.U.S.E.** (Japanese Union of Scientists and Engineers) was formed to develop more QC techniques for use by **managerial level staff**
1972-1977	The committee worked in conjunction with the original Basic Seven Tools
1977	A new set of methods called the **N7 (New Seven)** was published

1.3 What are the New Seven QC Tools?

The New 7 QC Tools are:

1. **Affinity Diagrams** (KJ Method) — Group chaotic ideas into themes
2. **Relations Diagrams** — Map complex cause-effect relationships
3. **Tree Diagrams** — Break objectives into actionable strategies
4. **Matrix Diagrams** — Show relationships between multiple factor sets
5. **Arrow Diagrams** — Plan and schedule tasks (like PERT/CPM)
6. **Process Decision Program Charts (PDPC)** — Plan for contingencies
7. **Matrix Data Analysis** — Prioritize using numerical scoring

2. Relationship: Basic 7 Tools vs New 7 Tools

Aspect	Basic 7 QC Tools	New 7 QC Tools
Data Type	Numerical / Quantitative data	Verbal / Qualitative data
Approach	Analytical approach	Planning & organizing approach
When Used	Define problem after collecting numerical data	Define problem before collecting numerical data
Purpose	Data analysis, process control, quality improvement	Generate ideas, formulate plans, organize information
Users	Shop floor workers, quality inspectors	Managers, planners, cross-functional teams

Analysis Point: The Basic 7 tools help you analyze what went wrong (reactive/diagnostic). The New 7 tools help you plan what to do next (proactive/strategic). Both sets are complementary — used together, they increase TQM effectiveness. Think of it as: Basic 7 = **diagnosis**, New 7 = **treatment planning**.

3. Tool 1: Affinity Diagrams (KJ Method)

"For Pinpointing the Problem in a Chaotic Situation and Generating Solution Strategies"

3.1 What is an Affinity Diagram?

> **Definition:** An Affinity Diagram is a **creative process used by a group to gather and organize verbal data** (ideas, opinions, issues) into natural groups based on their relationships. It was developed by Japanese anthropologist **Jiro Kawakita**, which is why it's also called the **KJ Method**.

- Organizes data into groups based on **natural relationship**
- Makes it feasible for **further analysis** and finding solutions
- Particularly useful when dealing with **chaotic, unstructured information**

3.2 How to Construct an Affinity Diagram (Group Method Approach)

> **Step-by-step construction:**
>
> 1. **Select a topic** — Define the problem or issue to explore
> 2. **Collect verbal data by brainstorming** — Every team member contributes ideas freely
> 3. **Discuss collected info** until everyone understands it thoroughly
> 4. **Write each item on a separate data card** (sticky notes work well)
> 5. **Spread out all cards on a table**
> 6. **Move data cards into groups of similar themes** (natural affinity for each other) — this is done silently, letting the groupings emerge naturally
> 7. **Create a title for each group** (this becomes the "Affinity Statement")
> 8. **Draw the affinity diagram**
> 9. **Present the results**

3.3 Completed Affinity Diagram Structure

Structure: Topic → Affinity Statements (group titles) → Data Cards (individual ideas). The diagram flows from a broad topic down to organized groups of related ideas.

3.4 Advantages of Affinity Diagrams

Advantages:

- Facilitates **breakthrough thinking** and stimulates fresh ideas
- Permits the problem to be **pinned down accurately**
- Ensures everyone **clearly recognizes** the problem
- **Incorporates opinions** of the entire group
- Fosters **team spirit**
- Raises everyone's **level of awareness**
- Encourages the group **into action**

Analysis Point: The Affinity Diagram is typically the first tool used in the New 7 sequence. When a problem is vague and no one knows where to start, the Affinity Diagram brings structure to chaos. The silent sorting step is key — it prevents dominant personalities from controlling the grouping and allows natural patterns to emerge.

4. Tool 2: Relations Diagrams (Interrelationship Diagrams)

4.1 What is a Relations Diagram?

Definition: A Relations Diagram is a tool for finding **solutions to problems that have complex causal relationships**. Unlike the fishbone diagram (which shows one-directional cause→effect), the Relations Diagram allows for **"multi-directional" thinking** rather than linear thinking. It is also known as an **Interrelationship Diagram**.

- Resolves **tangled issues** by untying the logical connections
- Allows for **multi-directional thinking** — causes can affect each other in complex ways
- Shows how multiple causes are **interrelated**, not just independent

4.2 How to Construct a Relations Diagram (Group Method)

Step-by-step construction:

1. **Express the problem** in the form of "Why isn't something happening?"
2. Each member **lists 5 causes** affecting the problem
3. **Write each item on a card**
4. **Discuss info collected** until everyone understands it thoroughly
5. **Move cards into similar groups**
6. Asking "why?", explore **cause-effect relationships** and divide cards into **primary, secondary, and tertiary causes**
7. **Connect all cards** by these relationships using arrows
8. **Further discuss** until all possible causes have been identified
9. **Review the whole diagram** looking for relationships among causes
10. **Connect all related groups**
11. **Complete the diagram**

4.3 Completed Relations Diagram

4.4 Advantages of Relations Diagrams

Advantages:

- Useful at the **planning stage** for obtaining perspective on overall situation
- Facilitates **consensus among team**
- Enables **priorities to be identified** accurately
- Makes the problem **recognizable** by clarifying the relationships among causes

Analysis Point: The Relations Diagram is like a more sophisticated version of the fishbone diagram. While the fishbone shows linear cause→effect, the Relations Diagram shows how causes interact with each other. For example, "lack of training" might cause "low quality" AND "low morale", and "low morale" might in turn worsen "training effectiveness" — these circular relationships are captured by the Relations Diagram but missed by a fishbone.

5. Tool 3: Tree Diagrams (Systematic Diagrams / Dendrograms)

"For Systematically Pursuing the Best Strategies for Attaining an Objective"

5.1 What is a Tree Diagram?

Definition: A Tree Diagram is a technique for **mapping out the full range of paths and tasks** that need to be done in order to achieve a **primary goal and related sub-goals**. It reveals in a simple way the magnitude of the problem and helps arrive at methods to achieve results. Also known as **Systematic Diagrams** or **Dendrograms**.

- Develops a **succession of strategies** for achieving objectives
- Reveals **methods to achieve the results**
- Breaks down a broad objective into increasingly specific, actionable levels

5.2 How to Construct a Tree Diagram

Step-by-step construction:

1. **Write the Relations Diagram topic** (Objective card)
2. **Identify constraints** on how the objective can be achieved
3. **Discuss means** of achieving the objective (primary means = first level strategy)
4. Take each primary mean and **write the objective for achieving it** (secondary means)
5. **Continue to expand** to the third level, fourth level, etc.
6. **Review each system of means** in both directions (from objective to means AND from means to objective)
7. **Add more cards** if needed
8. **Connect all levels**
9. **Complete the diagram**

5.3 Completed Tree Diagram

5.4 Advantages of Tree Diagrams

Advantages:

- **Systematic and logical approach** — less likely that items are omitted
- Facilitates **agreement among team**
- Makes complex objectives **manageable** by breaking them into smaller tasks

Analysis Point: The Tree Diagram takes the output of the Relations Diagram (identified causes/issues) and converts it into an **actionable plan**. Think of it as: Relations Diagram = "here's what's wrong and why" → Tree Diagram = "here's what we need to DO about it, broken into specific tasks." It's the bridge from analysis to action.

6. Tool 4: Matrix Diagrams

"For Clarifying Problems by Thinking Multidimensionally"

6.1 What is a Matrix Diagram?

> **Definition:** A Matrix Diagram is a versatile, graphical tool designed to facilitate the identification of **relationships between two or more sets of factors**. In strategic quality planning, factors are often interrelated — a change in one will affect others. The matrix helps us discover these relationships and trade-offs in a structured, visual way.

- Consists of a **two-dimensional array** to determine location and nature of the problem
- Discovers key ideas by **relationships represented by the cells** in the matrix
- Helps address **trade-offs** in strategic planning

6.2 When to Apply a Matrix Diagram

Applied when the interrelationship of the following needs to be presented:

- **Effects & causes**
- **Objectives & methods**
- **People & functions**

The interrelationships need to be characterized as:

- **Exists or does not exist**
- **Strong, medium, weak**
- **Type** (positive or negative)

6.3 How to Construct a Matrix Diagram

Step-by-step construction:

1. **Write final-level means** from the Tree Diagram forming the **vertical axis**
2. Write in **Evaluation categories** (efficacy, practicability, and rank) on the **horizontal axis**
3. Examine final-level means to identify **who will implement them**
4. **Write names** along the horizontal axis
5. Label group of columns as **"Responsibilities"**
6. Label right-hand end of horizontal axis as **"Remarks"**
7. Examine each cell and insert the appropriate **symbol**:
    - Efficacy: O = good, △ = satisfactory, X = none
    - Practicability: O = good, △ = satisfactory, X = none
8. **Determine score** for each combination of symbols, record in rank column
9. Examine cells under Responsibility columns — insert **double-circle for Principal** and **single-circle for Subsidiary**
10. Fill out **remarks** column
11. **Complete the diagram**

6.4 Completed Matrix Diagram

6.5 Advantages of Matrix Diagrams

Advantages:

- Enable data or ideas based on **extensive experience**
- **Clarifies relationships** among different elements
- Makes **overall structure of problem** immediately obvious

Analysis Point: The Matrix Diagram takes the actionable items from the Tree Diagram and helps you **evaluate and assign** them. It answers: "Which strategies are most effective? Who is responsible for each?" This is where planning meets execution. The symbol-based evaluation (O, △, X) makes it quick to scan and prioritize.

7. Tool 5: Arrow Diagrams

"For Working Out Optimal Schedules and Controlling Them Effectively"

7.1 What is an Arrow Diagram?

> **Definition:** An Arrow Diagram shows **relationships among tasks needed to implement a plan**. It is a network technique using **nodes for events** and **arrows for activities**. It is essentially the same concept used in **PERT (Program Evaluation and Review Technique)** and **CPM (Critical Path Method)**.

- Shows the sequence and dependencies between tasks
- Identifies which tasks can be done in **parallel** and which must be **sequential**
- Helps find the **critical path** — the longest sequence of dependent tasks that determines the minimum project duration

7.2 How to Construct an Arrow Diagram

Step-by-step construction:

1. From strategies, **select one** (Objective of Arrow Diagram)
2. **Identify constraints** to the objective
3. **List all activities** necessary to achieve the objective
4. **Write all essential activities** on separate cards
5. **Organize cards in sequential order** of activities
6. **Remove any duplicate** activities
7. **Review order of activities**, find sequence with the greatest amount of activities
8. **Arrange parallel activities** (activities that can happen simultaneously)
9. **Examine path**, number nodes in sequence from left to right
10. **Record names** and other necessary information
11. **Complete the diagram**

7.3 Completed Arrow Diagram

7.4 Advantages of Arrow Diagrams

Advantages:

- Allows overall task to be **viewed and potential problems identified** before work starts
- Leads to **discovery of possible improvements**
- Makes it easy to **monitor progress** of work
- Deals promptly with **changes to plan**
- Improves **communication among team**
- Promotes **understanding and agreement** among group

Analysis Point: If you've studied Project Management, PERT, or CPM, the Arrow Diagram will feel familiar. The key quality management insight is that it connects quality planning to quality execution — you don't just identify what needs to be done (Tree Diagram), you also determine when and in what order it needs to happen. The critical path identifies where delays would impact the entire project.

8. Tool 6: Process Decision Program Charts (PDPC)

"For Producing the Desired Result from Many Possible Outcomes"

8.1 What is a PDPC?

> **Definition:** The Process Decision Program Chart (PDPC) is a very useful and powerful method to **overcome a problem or achieve a goal**. With PDPC, we can map out all conceivable **events or contingencies** that can occur in the implementation stage and also find **feasible counter measures** to overcome these problems.

- Used to **plan various contingencies** — "what if?" planning
- Used for getting activities **back on track** when things go wrong
- **Steers events** in the required direction if unanticipated problems occur
- Finds **feasible counter measures** to overcome problems

8.2 How to Construct a PDPC

Step-by-step construction:

1. **Select a highly effective, but difficult strategy** from the Tree Diagram
2. **Decide on a goal** (most desirable outcome)
3. **Identify existing situation** (starting point)
4. **Identify constraints** of the objective
5. **List activities** to reach the goal AND **potential problems** with each activity
6. **Review list** — add extra activities or problems not previously thought of
7. **Prepare contingency plan** for each step — review what action is needed if a step is NOT achieved
8. **Examine carefully** to check for inconsistencies and ensure all important factors are included
9. Examine to make sure all **contingency plans are adequate**
10. **Complete the diagram**

8.3 Completed PDPC

How to read the PDPC: Start at the top. At each activity, ask "Was it successful?" If YES, proceed to the next step. If NO, follow the contingency path (alternative action). The goal is to reach the GOAL box regardless of which path you take. Every "NO" has a planned response.

8.4 Advantages of PDPC

Advantages:

- Facilitates **forecasting**
- Uses past experience to **anticipate contingencies**
- Enables problems to be **pinpointed**
- Illustrates how events will be **directed to successful conclusion**
- Enables those involved to understand **decision-makers' intentions**

Analysis Point: PDPC is like a "Plan B generator." Most plans assume everything will go perfectly. PDPC recognizes that things WILL go wrong and pre-plans responses. This is particularly valuable in quality management where process failures can be costly. Think of it as combining a flowchart with risk management.

9. Tool 7: Matrix Data Analysis (Prioritization Grid)

"Principal Component Analysis"

9.1 What is Matrix Data Analysis?

> **Definition:** Matrix Data Analysis is a technique that **quantifies and arranges data presented in a Matrix Diagram**. It is based solely on **numerical data** (making it the only New 7 tool that uses numbers). It finds indicators that differentiate and attempts to **clarify large amounts of information**. It is also known as **Principal Component Analysis**.

- Can be used in various fields: **market surveys, new product planning, process analysis**
- Used when Matrix Diagram does not give **sufficient information**
- Useful as a **Prioritization Grid** — helps rank alternatives based on weighted criteria

9.2 How to Construct a Prioritization Grid

Step-by-step construction:

1. **Determine your goal**, your alternatives, and your criteria for decision
2. **Place selection criteria** in order of importance
3. **Apply percentage weight** to each option (all weights should add up to 1)
4. **Sum individual ratings** to establish overall ranking (divide by number of options for average ranking)
5. **Rank order each option** with respect to each criterion (average the rankings and apply a completed ranking)
6. **Multiply weight by associated rank** in the matrix (e.g., 4 is best, 1 is worst)
7. The result is the **Importance Score**
8. **Add up Importance Scores** for each option
9. **Rank order the alternatives** according to importance

9.3 Completed Prioritization Grid

Analysis Point: Matrix Data Analysis is the most mathematical of the New 7 tools. It converts subjective judgments into objective scores. The Prioritization Grid is especially useful when you have multiple alternatives (designs, strategies, vendors) and need to choose the best one based on multiple weighted criteria. It removes bias by forcing structured, numerical evaluation.

10. Summary — Benefits, Roles & Quick Reference

10.1 Benefits of the New Seven QC Tools

1. **Provide training in thinking** — develop structured problem-solving skills
2. **Raise people's problem-solving confidence** — give teams systematic methods
3. **Increase people's ability to predict future events** — through contingency planning (PDPC) and scheduling (Arrow Diagrams)

10.2 Roles of the New Seven QC Tools

1. **Express verbal data diagrammatically** — make the invisible visible
2. **Make information visible** — transform abstract ideas into concrete diagrams
3. **Organize information intelligibly** — structure chaos
4. **Clarify overall picture and fine details** — both big-picture and granular view
5. **Get more people involved** — collaborative, team-based methods

10.3 Quick Reference Table — All 7 New QC Tools

Tool	Purpose	Key Question It Answers	Data Type
1. Affinity Diagram	Organize chaotic ideas into themes	"What are the key themes in this mess?"	Verbal
2. Relations Diagram	Map complex cause-effect networks	"How do these causes interrelate?"	Verbal
3. Tree Diagram	Break objectives into strategies/tasks	"HOW do we achieve this goal?"	Verbal
4. Matrix Diagram	Evaluate & assign responsibilities	"Who does what? How effective is each?"	Verbal + Symbols
5. Arrow Diagram	Schedule tasks & find critical path	"WHEN and in what order?"	Verbal
6. PDPC	Plan contingencies	"What if something goes wrong?"	Verbal
7. Matrix Data Analysis	Prioritize alternatives numerically	"Which option is objectively best?"	Numerical

10.4 How the New 7 Tools Flow Together

Typical workflow using the New 7 QC Tools:

Affinity Diagram (organize the chaos) → **Relations Diagram** (find the root causes and their connections) → **Tree Diagram** (break down objectives into actionable strategies) → **Matrix Diagram** (evaluate strategies and assign responsibilities) → **Arrow Diagram** (schedule the implementation) → **PDPC** (plan for contingencies) → **Matrix Data Analysis** (prioritize when numerical comparison is needed)

Exam Tip: Remember the distinction: **Basic 7 = Numerical data, Analytical** (for shop floor). **New 7 = Verbal data, Planning** (for management). Only Matrix Data Analysis uses numbers. Know the purpose of each tool, its construction steps, and at least 3 advantages. The flow from Affinity → Relations → Tree → Matrix → Arrow → PDPC is a common exam question.